ASSIGNMENT 1 – TASK 2: Choose the Data Representation

Bloom's Level: K4 – Analyse | Marks: 3

Data	Chosen representation	Reason
Floor index, slot index, counts	int	Whole-number indexes; validate before array access.
Registration number	String	Contains letters and digits; preserve leading zeros.
Entry and exit	Instant or LocalDateTime	Use Instant for elapsed duration; LocalDateTime for explicitly local campus rules.
Duration	long minutes or Duration	Avoid overflow and unit confusion.
Parking fee	BigDecimal or long paise	Currency needs exact decimal/minor-unit handling.
Slots per floor	String[][] for the supplied template	Jagged array allows each floor to have a different number of slots.
Tickets	ArrayList<ParkingTicket>	Stores tickets in order of creation; search is linear unless indexed.
Ticket number	long from TicketNumberGenerator	Avoid reuse and provide a wide range.

A jagged array is appropriate because slots[i] can have a different length for each floor. For the provided beginner-level examples, double is used only to keep arithmetic easy to read; this is acceptable for demonstration output but not ideal for real currency because binary floating-point may round unexpectedly. Production billing should use BigDecimal or integer paise.

The task also asks about ArrayList<String[]> for the slots. Here the list holds one String[] per floor: the list index is the floor and the array index is the slot. list.get(f)[s] locates a slot in O(1) time, floors may have different lengths, and the space is O(F + N) plus the list's own overhead. With a single flat array, unequal floors need prefix offsets (the total capacity of earlier floors) to map floor and slot to an index. Input can be read with Scanner for interactive use, or from command-line arguments, which arrive as Strings and must be parsed and validated.

Integer division risk

```
int billedHours = (minutes + 59) / 60; // ceiling division for non-negative minutes
double fee = billedHours * hourlyRate;
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For 61 minutes, 61 / 60 using integer division gives 1 and undercharges if every started hour is billed. Ceiling division gives 2. Reject negative minutes before this calculation.

One-dimensional alternative

A flattened array can map equal floor capacities using index = floor * slotsPerFloor + slot. With unequal capacities, use prefix offsets (the total capacity of earlier floors) or a mapping table. The jagged String[][] is clearer for this assignment and makes release by floor/slot O(1).